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How to evaluate a model trained via SSL?

Less work

More work

kNNLinear classifierFine-tune all layers

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Unified theories of self-supervised learning

- ▶ The only way to formalize & understand all model types
- ▶ Gives low energy to compatible pairs of  $x$  and  $y$
- ▶ Gives higher energy to incompatible pairs

line of space

Energy Landscape

$\hat{y} = \operatorname{argmin}_y F(x, y)$

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Unified theories of self-supervised learning

- ▶ A flexible energy surface can take any shape.
- ▶ We need a loss function that shapes the energy surface so that:
  - ▶ Data points have low energies.
  - ▶ Points outside the regions of high data density have higher energies.

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Unified theories of self-supervised learning

6 CONCLUSION

Through an analysis of their criteria, we were able to show that simple contrastive and dimension-contrastive methods have learning objectives that are closely related, as they fundamentally maximize the mutual information of the observations. This suggests a certain duality in the behavior of such methods, which we studied empirically. Through the lens of variational VCBs, we were able to study regular deep chains, as self-supervised loss functions and show their lack of impact on performance, significantly improving the robustness of embedding dimension VCBs along the way. Motivated by our theoretical findings, we performed graph representation learning on GraphSAGE and were able to show its performance gap with VCBs. We also showed that the intermediate energy also was able to improve our results with VCBs. The latter method is therefore relevant to the problem of the behavior of results. We expect that our results will help related theoretical works in self-supervised learning as a wider family of methods, as well as help analyze by deriving criteria that are easier to work with. We also expect that our findings will help relevant practitioners learn on contrastive and non-contrastive learning. If our findings are corroborated from this work, it is our conviction that contrastive and regularized contrastive methods are the state of the state-of-the-art. Finally, perhaps the most important message of this work is to show that different SSL methods can be unified.

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Unified theories of self-supervised learning

Matrix of representations for a Batch of Samples

variables

samples

- ▶ Sample Contrastive Methods:
  - ▶ Make the row of the matrix as different from each other as possible
  - ▶ Requires a large number of rows
  - ▶ Don't work in high dimension
- ▶ Dimension Contrastive Methods:
  - ▶ Make the column as different from each other as possible
  - ▶ Requires a small number of rows
  - ▶ Don't work for large batches

▶ Equivalence

Simola VLJ 2022  
Arxiv:2208.02574

On the duality between contrastive and non-contrastive self-supervised learning

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Resources

2023 ICML tutorial on SSL: <https://icml.cc/virtual/2023/tutorial/21552>

A Cookbook of Self-Supervised Learning: <https://arxiv.org/abs/2304.12210>

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